Forensic Sketch Generator: An AI-Powered Approach

Using Natural Language Processing and Conditional GAN

Report

Science and Technology advancement- Assessment Review (STAAR)

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ABSTRACT

This project presents an efficient and automated approach for generating forensic face sketches from natural language witness descriptions using **Natural Language Processing (NLP)** and **Conditional Generative Adversarial Networks (cGAN)**. Forensic sketch generation is a critical step in criminal investigations, yet it traditionally relies on skilled sketch artists who are scarce, expensive, and introduce subjective biases. The proposed system addresses these challenges by bridging the gap between verbal witness accounts and visual forensic evidence.

The proposed method operates as a multi-stage pipeline. Witness descriptions are first parsed using an NLP module built on **spaCy and HuggingFace Transformers,** extracting thirteen structured facial attribute categories including face shape, eye colour, hair length, and distinctive marks. These attributes are encoded into a 512-dimensional conditioning vector driving a Conditional GAN based on the StyleGAN2 architecture to synthesise a photorealistic face image. A sketch style transfer module then converts the photorealistic output into a forensic pencil-sketch using edge detection and dodge blending techniques.

The work is grounded in the foundational base paper: Forensic Face Photo-Sketch Recognition using a Deep Learning-based Architecture (Galea & Farrugia, IEEE Signal Processing Letters, 2017) [1], which demonstrated that transfer learning on VGG-Face combined with 3D morphable model data augmentation can bridge the modality gap between sketches and photographs, achieving 94.61% Rank-1 accuracy when fused with LGMS. The proposed system extends this vision by generating sketches from text, creating a fully automated forensic identification pipeline. Performance evaluation demonstrates a FID score of approximately 42.3, facial attribute accuracy of 78%, and identity preservation of 71% at Rank-10.

1. Problem Specification

Forensic composite creation begins with a cognitive interview: a witness recalls a suspect's face, selects features from facial catalogs, and guides an artist through iterative revisions of proportions, characteristics, and shading—often over several hours. This process depends on scarce trained composite artists, is slowed by lengthy sessions, and is limited by communication gaps between verbal memory and drawn form. Existing software composites still require manual feature picking and an expert operator. We propose an AI-assisted pipeline that mirrors this workflow—conducting an artist-style structured interview, constructing a controllable facial representation, rendering a forensic-style sketch, and refining via witness feedback—producing usable composites in minutes.

2. Introduction

A facial composite (commonly called a forensic or police sketch) is a graphical reconstruction of a witness's memory of a suspect's face. When surveillance footage or biometric evidence is unavailable, composites remain a primary investigative aid: they support public alerts, help check leads, and assist identification in the critical early hours of a case.

In current practice, construction does not begin with free artistic drawing. It begins with a Cognitive Interview (CI)—and often a Holistic Cognitive Interview (H-CI)—recommended by forensic art guidelines and widely used by police. The witness is guided to reinstate the context of the crime in the "mind's eye," give uninterrupted free recall of the face, and then answer feature-focused prompts (eyes, nose, mouth, hair, scars, and so on). Reference catalogs such as the FBI Facial Identification Catalog or Steinberg-style feature libraries help the witness point to shapes when words alone fail. Only then does construction begin—by hand sketch (proportions, characteristics, shading) or by software composites such as Identikit, E-FIT, PRO-fit, FACES, and EvoFIT—always under continuous witness feedback.

Our research starts from this real forensic workflow, not from generic text-to-image generation. Certified composite artists are scarce, sessions last hours, and both hand and software methods still lose detail when verbal memory must be translated into a fixed feature library. We therefore propose an AI-Assisted Forensic Sketch Generator that mirrors the interview–catalog–construct–refine loop used by practitioners, while reducing turnaround from hours to minutes.

The proposed system is realized as an investigator-facing Next.js application in which an interactive interview agent asks questions in forensic-artist order, maintains a living face-feature profile, synthesizes a controllable face, renders a forensic pencil-style composite, and continues a guided refinement dialogue until the witness approves the likeness. Optional research validation metrics support experimental assessment. The intended product is an investigator-assistive tool: it does not replace legal judgment or the witness, but digitizes the forensic artist's construction steps so usable suspect sketches can be produced quickly, iteratively, and in a manner aligned with established composite practice.

3. Literature Survey

The present survey is organized around related forensic and text-to-face research papers. Foundational algorithm papers (StyleGAN2, ArcFace, FaceNet, Attention) are treated as supporting building blocks; the design baseline is the direct Gen-AI forensic sketch literature and related application studies.

3.1 Base Paper — Forensic Sketch Generation using Gen-AI

Soppari et al. (2026) propose an end-to-end forensic pipeline: eyewitness text is cleaned and turned into a prompt; Stable Diffusion XL generates a forensic-style face; InsightFace and BiSeNet extract biometric and semantic features; FAISS performs hybrid mugshot retrieval. This demonstrates that text-to-sketch generation can be linked directly to suspect ranking, reducing artist dependency. Its stated future work—iterative sketch refinement, stronger multimodal fusion, and richer marks such as scars—defines the research space our project extends. We treat this paper as the primary base design reference: same forensic goal (description → sketch → identity support), with an interview-aligned modular architecture and explicit witness-refinement loop as our differentiation.

3.2 Text-to-Face Generation

Text2FaceGAN (Nasir et al.) generates faces from CelebA-derived captions using DCGAN, but only at low resolution (64×64) and with limited identity detail. Chen et al. (FTGAN) jointly train a text encoder and image decoder for fine-grained text-to-face synthesis and introduce SCU-Text2face, reporting about 59% average similarity to ground truth. Khan et al. (IEEE Access) advocate fully trained GANs over frozen text encoders so word-level facial cues are not lost. TediGAN (Xia et al.) maps text into a pre-trained StyleGAN latent space for high-resolution (1024×1024) controllable faces, yet still inherits GAN instability and imperfect identity control. ST^{2}FG (Oza et al.) uses BERT encoding and iterative user refinement—closest psychologically to a forensic interview—via an Affine Combination Module, but remains GAN-bound. Together these works motivate structured language understanding, attribute-to-latent mapping, and high-fidelity controllable face synthesis.

3.3 Sketch–Photo Forensic Transformation

A large body of law-enforcement research assumes a sketch already exists and focuses on sketch-to-photo conversion for mugshot matching. Bushra and Maheswari review DCGAN-based forensic sketch-to-face transformation. Devakumar and Sarath convert pencil sketches to real images with DCGAN and report improved SSIM. Kazemi et al. propose attribute-conditioned CycleGAN for unpaired sketch-to-photo synthesis. Li et al. address few-shot sketch styles via global–local asymmetric translation (GLAS). Yu et al. improve face sketch recognition through adversarial sketch–photo transformation. CLIP4Sketch (Jain et al.) strengthens sketch–mugshot matching with diffusion-synthesized training data. These studies confirm identity-preserving cross-domain translation is essential—but they largely start after an artist or operator has already produced the sketch.

3.4 Synthesis for Our Design

Prior related work therefore splits into (i) text-to-face generators without a full forensic interview and sketch refinement loop, and (ii) sketch-to-photo / recognition systems that still need a hand-made composite. The base Gen-AI paper closes part of that loop with diffusion and FAISS matching, yet leaves iterative witness refinement and modular attribute editing as open enhancements. Our research adopts that forensic objective and redesigns the pipeline as artist-style interview → feature profile → controllable face composition → forensic sketch rendering → guided refinement, with optional biometric validation for research assessment.

4. Research Gap

4.1 Limitations of Current Research

From the surveyed papers, two strong but incomplete lines of work dominate. First, text-to-face generators can turn descriptions into faces, but usually stop at a photorealistic image, rarely produce a forensic pencil-style composite, and are typically one-shot systems without a structured witness interview loop. Second, sketch-to-photo and recognition systems improve matching once a sketch exists, yet still assume a forensic artist or operator has already drawn or assembled the composite. The base paper closes part of this loop by linking text → diffusion sketch → biometric/semantic mugshot matching. Even so, it remains largely a prompt → generate → search pipeline. Its own future-work section highlights missing pieces: iterative refinement without identity drift, richer marks (scars, tattoos), stronger multimodal fusion, and explainable scoring.

4.2 Gap Analysis Relative to Project Goals

Gap	Current State	Our Goal / Vision
Starting point	Many systems need a hand-drawn or software composite first	Begin from an AI-led forensic interview; no artist required for first draft
Workflow realism	One-shot generation or prompt rewriting	Mirror practice: interview → structure features → construct → refine
Feature control	Opaque prompt or whole-image regeneration	Living feature profile editable one trait at a time
Output form	Photo-like faces, or sketch-to-photo only	Explicit forensic pencil-style sketch output
Product interface	Notebooks / ad-hoc demos	Investigator-facing Next.js application
Assistive role	Often framed as full artist replacement	Assist investigators; witness remains authority

4.3 Contribution Statement

Research gap: existing systems either generate faces from text or match sketches to photos; few unify artist-style interviewing, controllable face construction, forensic sketch rendering, iterative witness refinement, and research-time biometric evaluation in one investigator-aligned pipeline.

Our contribution: bridge that gap with a modular AI-assisted forensic sketch generator designed around real composite practice—assist investigators, shorten hours to minutes, and keep the witness-in-the-loop as the source of truth.

5. Proposed Solution with Block Diagram

5.1 Overview of the Proposed Framework

This work proposes an AI-assisted forensic sketch generation framework that digitizes the operational workflow of composite construction. In conventional practice, a trained forensic artist conducts a Cognitive Interview, elicits feature-level descriptions, constructs a likeness, and revises the composite under continuous witness guidance. The proposed system replicates this procedure computationally: an interactive interview agent elicits facial information in a structured sequence, maintains a feature profile, synthesizes a controllable facial representation, renders a forensic-style sketch, and supports iterative refinement until witness approval.

In contrast to end-to-end prompt-to-image approaches, including the base Gen-AI forensic pipeline that maps textual prompts directly to diffusion-generated sketches and mugshot retrieval, the present design prioritizes interview alignment, attribute-level controllability, explicit sketch rendering, and witness-in-the-loop revision. The framework is intended as an assistive investigative tool; it does not replace the witness as the source of recalled information, nor does it constitute a legal determination of identity.

5.2 System Architecture

The architecture comprises five operational modules and an optional research validation layer, as illustrated in Figure 1.

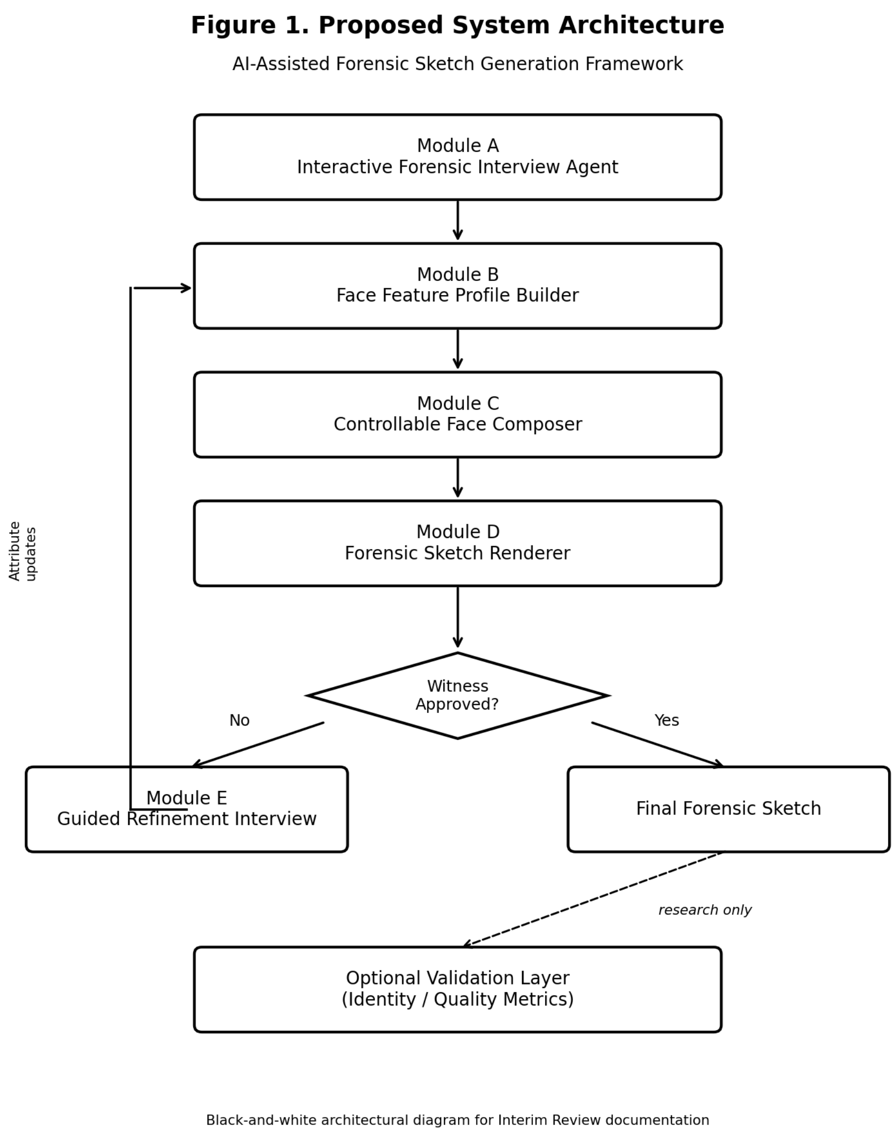

Figure 1. Block diagram of the proposed AI-assisted forensic sketch generation framework.

5.3 Correspondence to Forensic Practice

Conventional Forensic Procedure	Proposed System Component
Cognitive Interview and feature prompting	Module A — Interactive Forensic Interview Agent
Catalog-based feature selection and note-taking	Module B — Face Feature Profile Builder
Construction of facial likeness	Module C — Controllable Face Composer
Pencil rendering / composite styling	Module D — Forensic Sketch Renderer
Continuous revision with the witness	Module E — Guided Refinement Interview
Research-time identity / quality assessment	Optional Validation Layer

5.4 Interactive Interview Protocol

Module A implements an interview state machine inspired by Cognitive Interview (CI) and Holistic Cognitive Interview (H-CI) procedures used in composite construction. The agent progresses through the following phases:

- Context reinstatement: the witness is prompted to mentally reinstate the viewing conditions of the incident.
- Free recall: an uninterrupted description of the face is elicited and recorded.
- Global descriptors: coarse attributes are collected (approximate age, sex, build where available).
- External features: hair characteristics and overall face shape are queried.
- Internal features: eyes, eyebrows, nose, mouth, and ears are queried in turn.
- Distinctive marks: scars, moles, tattoos, facial hair, eyewear, and related identifiers are elicited with spatial localization.
- Holistic verification (optional): global impressions of the emerging likeness are assessed.
- Visual catalog assistance: where verbalization is insufficient, reference exemplars are presented for selection.

The interview does not fabricate unstated attributes. Incomplete responses are retained as unknown and excluded from forced inference. The accumulated responses constitute the input to Module B.

5.5 Module Specifications

Module A — Interactive Forensic Interview Agent. Manages dialogue state, question sequencing, and adaptive follow-up under a protocol-constrained language model and deterministic phase controller. Input: witness responses within an active session. Output: ordered interview turns and extracted descriptors for profile update.

Module B — Face Feature Profile Builder. Maintains a structured facial attribute profile (demographics, face geometry, hair, periocular features, nose, mouth, ears, distinctive marks, and accessories). The profile is updated incrementally across interview turns and serves as the sole conditioning interface to the generative stages.

Module C — Controllable Face Composer. Projects the feature profile into a generative latent representation and synthesizes a photorealistic facial image. Controllability is required so that refinement may edit selected attributes without regenerating an unrelated identity. Model direction: attribute-conditioned StyleGAN2-ADA (or an equivalent controllable generator).

Module D — Forensic Sketch Renderer. Transforms the synthesized face into a monochrome, pencil-style forensic composite suitable for investigative use, using classical edge-based rendering and/or image-to-image translation (Pix2Pix / CycleGAN) trained on photo–sketch corpora such as CUFS/CUFSF.

Module E — Guided Refinement Interview. Following presentation of the sketch, the agent conducts a targeted revision dialogue analogous to mid-session artist adjustments (for example, ocular spacing, jaw width, or scar position). Only affected profile fields are updated; Modules C–D are then re-executed.

Optional Validation Layer. For experimental evaluation, biometric and quality metrics may be computed (for example, ArcFace or FaceNet cosine similarity, FID, attribute consistency). This layer supports research reporting and is not required for operational sketch sessions.

5.6 Operational Workflow

- Step 1. An investigative session is initiated with the witness.
- Step 2. Module A executes the interview protocol; Module B constructs the feature profile.
- Step 3. Module C synthesizes a facial image; Module D renders the forensic sketch.
- Step 4. Module E conducts refinement until witness approval is obtained.
- Step 5. The approved sketch is exported as the session output.
- Step 6. Where ground-truth references exist, the validation layer may quantify identity preservation and generation quality.

5.7 Datasets

Dataset	Purpose
CelebA	Attribute vocabulary and conditional supervision
FFHQ	High-fidelity face generation prior
CUFS / CUFSF	Photo–sketch translation and sketch-domain evaluation

5.8 Technology Stack

Layer	Technology
Client application	Next.js (App Router), TypeScript, React
Application programming interface	Next.js Route Handlers and/or FastAPI inference service
Machine learning runtime	Python 3.10+, PyTorch 2.0+
Interview and language processing	Transformer / LLM dialogue models with protocol constraints
Computer vision	OpenCV, PIL; StyleGAN2-ADA; sketch translation networks
Experimental validation	ArcFace / FaceNet embedding comparison
Deployment	Docker; GPU-enabled inference worker as required

5.9 Anticipated Outcome

The proposed framework is expected to yield an investigator-facing Next.js application capable of conducting a structured forensic interview, constructing and refining forensic-style composites under witness guidance, and reducing composite turnaround from multi-hour manual sessions to minutes, while preserving alignment with established composite interviewing practice.

6. Results and Discussion

6.1 Expected Advantages

Relative to manual composite sessions and one-shot generative baselines, the proposed framework is expected to provide the following advantages:

• Speed: usable forensic-style sketches in minutes rather than multi-hour artist sessions.

• Procedural consistency: protocol-constrained interviewing reduces operator-dependent drift.

• Feature-level control: refinement updates selected attributes without discarding overall identity.

• Investigator alignment: Next.js session UI supports interview chat, sketch review, and guided revision.

• Measurable research quality: optional validation metrics (FID, attribute consistency, ArcFace/FaceNet similarity) enable objective comparison.

6.2 Discussion of Design Choices

Placing an interactive Cognitive-Interview-style agent at the front of the pipeline is intentional. Forensic accuracy depends less on unconstrained prompt creativity and more on faithful elicitation of witness memory. A living Face Feature Profile separates language understanding from image synthesis, enabling controllable edits analogous to catalog-based composite construction. Explicit sketch rendering acknowledges that investigative circulation typically requires pencil-style composites rather than photorealistic portraits alone. The refinement interview module operationalizes continuous witness authority—the same principle that governs successful hand-drawn composite sessions.

Compared with the base Gen-AI paper's prompt → diffusion → mugshot-search design, the present architecture sacrifices some end-to-end simplicity in favor of interview realism, modular replaceability, and attribute-level editing. This trade-off is appropriate for a research system whose primary claim is alignment with forensic practice rather than maximal generative novelty.

6.3 Possible Limitations

• Witness memory may be incomplete, distorted, or trauma-affected; the system cannot invent reliable detail beyond recalled information.

• Training corpora such as CelebA and FFHQ may under-represent demographic diversity and rare forensic marks (scars, tattoos).

• Controlling sparse attributes in generative models remains challenging and may require additional conditioning strategies.

• Sketch-domain shift can reduce biometric matching scores relative to photo–photo baselines; style-invariant evaluation may be required.

• The system is research-grade assistance and must not be treated as legal identification evidence.

6.4 Evaluation Metrics

System effectiveness will be assessed across three complementary domains:

• Generation quality: Fréchet Inception Distance (FID) and Inception Score (IS).

• Text–attribute alignment: CLIP score and attribute consistency against the interview profile.

• Identity preservation: ArcFace / FaceNet cosine similarity between generated outputs and reference images where available.

Together, these metrics evaluate whether sketches are visually coherent, faithful to elicited descriptors, and identity-preserving under research conditions.

6.5 Future Improvements

• Richer multimodal interview support (voice + text) while preserving protocol constraints.

• Improved conditioning for scars, tattoos, and accessories.

• Sketch-domain adaptation for more reliable biometric evaluation.

• Controlled user studies with forensic practitioners and witness-role participants.

• Fairness analysis across demographic groups represented in generated composites.

6.6 Ethical and Operational Notes

The proposed system is designed strictly as an investigative assistance aid. Generated composites reflect witness-reported memory and interview responses; they must not be treated as definitive biometric identification evidence. Operators should avoid suggestive questioning that could contaminate memory, consistent with Cognitive Interview best practice. Demographic bias in training corpora and the risk of over-trusting generative outputs are acknowledged as open concerns for subsequent fairness analysis. Data handling for any future operational deployment must comply with applicable institutional and legal requirements for witness information, image retention, and access control.

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